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DIGITAL PASSWORD LOCK SYSTEM

Neng-232 Lab Project

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Section 251

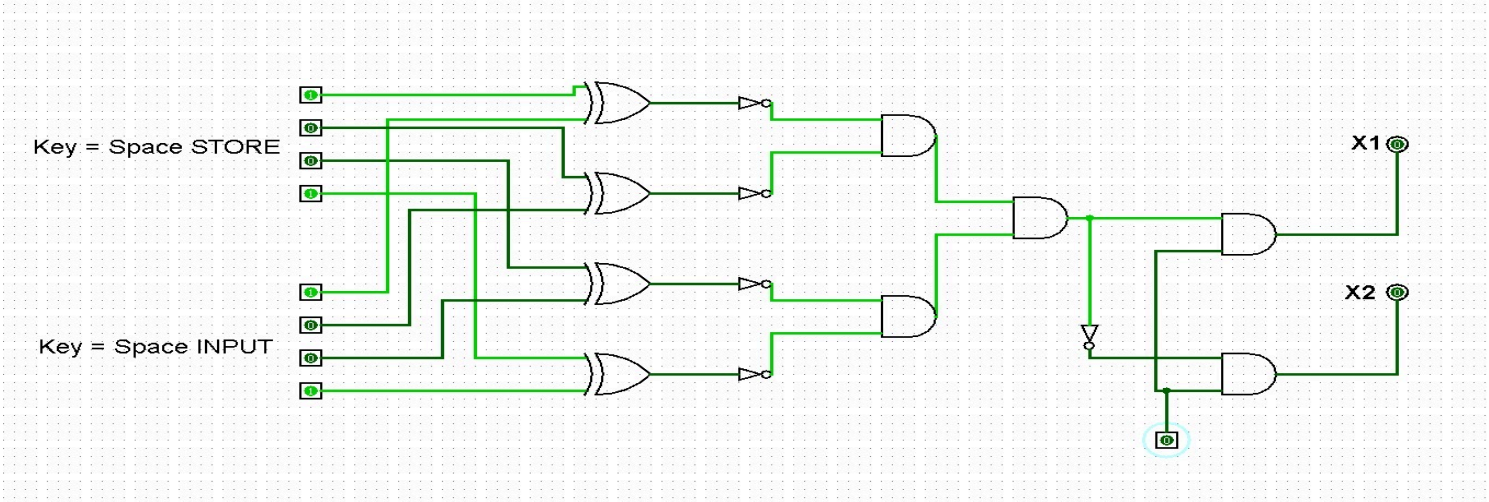
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Chapter One

Project Idea + Introduction to Logic Gates

Project Idea



This project is a digital password-lock system built completely using basic logic gates. The main idea is to create a hardware circuit that compares a stored 4-bit password with a 4-bit input entered by the user. The system checks each bit of the password using XOR, NOT, and AND gates to determine whether the entered password is correct or not.

The main goal of this project is to design a simple, reliable, and low-cost password-checking circuit that operates purely with logic gates, and to help understand how digital comparisons and decision-making can be implemented at the hardware level without any programming.

Overall, the project demonstrates how digital logic gates can be used to build a simple and functional password verification system without using any microcontrollers or memory components.

Introduction to Logic Gates

Logic gates are the basic building blocks of digital circuits. They take one or more binary inputs (0 or 1) and produce a single binary output based on a specific logical rule. Logic gates are used everywhere in digital systems such as computers, microprocessors, memory units, digital locks, and control circuits. By combining gates, we can design systems that make decisions, store information, compare data, and control outputs.

Logic gates are essential because they allow hardware to perform logical operations like checking conditions, comparing values, and executing instructions — all using electricity.

- **Gates Used in This Project:**

1- XOR Gate:

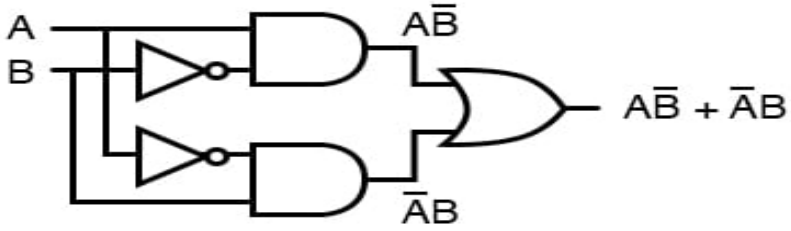


A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

The XOR gate is a logic gate where the output is 1 if one – and only one – of its inputs are 1. XOR stands for Exclusive-OR. The logic or Boolean expression for an XOR gate is $A \oplus B$ = Q. Which means that "If A And B are different from each other, then Q is true". XOR gates are usually found with just two inputs. If either of the two inputs is 1, while the other is 0, then the output becomes 1. Said in a different way, the output is 1 (High) only when its two inputs are not the same.



. . . is equivalent to . .



$A \oplus B = A\bar{B} + \bar{A}B$

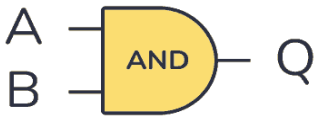
2- NOT Gate (Inverter):



A	Q
0	1
1	0

A NOT gate (or inverter) is a logic gate where the output is the opposite of the input. So we can say that the output is NOT the same as the input. It's often called an *inverter* since it inverts the input. The schematic symbol for an inverter is like a buffer, just with a circle at the output to indicate that it's an inverted version of the input. The logic or Boolean expression for a NOT gate is $Q = \bar{A}$. Which means that "Q is the opposite of A". A NOT gate can only have one input. And its truth table is pretty simple since there are only two possible states; the input being 1 (High) or the input being 0 (Low). In this digital circuit, combining a NOT gate with an XOR gate result in an XNOR gate. An XNOR gate output is 1 only if both its inputs are equal. So if the inputs A and B are both 1 or both 2, the output Q will be 1.

3- AND Gate:



A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

An AND gate is a logic gate where the output 1 only if all its inputs are 1. So if the inputs A **AND** B are 1, the output Q will also be 1, or it will be 0 otherwise. The logic or Boolean expression for an AND gate is $Q = A \cdot B$. Which means that "If A and B are true, then Q is true. AND gates can have more than two inputs. But no matter how many inputs it has, it will only give out a 1 if all its inputs are 1. As soon as one of the inputs is 0, the output will be 0 too.

Chapter Two

Project Implementation Steps

The digital circuit consists of:

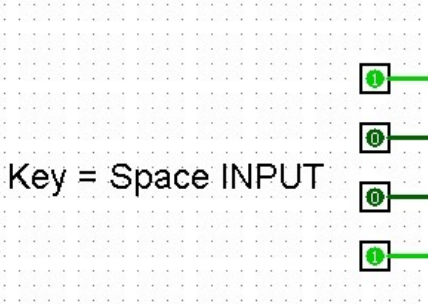
1- Space STORE Switches: Used to store 4-bit password.
2- Space INPUT Switches: Used to take 4-bit input password from user.
3- Input Enable Switch: Enables the System when it equals 1.
4- First Output (X1): Valid password checker.
5- Second Output (X2): Invalid password checker.
6- XOR Gates.
7- NOT (Inverter) Gates.
8- AND Gates.

- **Space STORE switches**



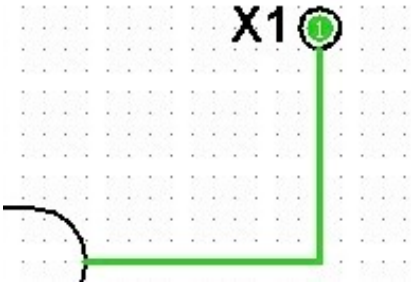
These switches represent the stored password, which is the password chosen by the designers of the system. These four bits are the reference password that the circuit uses for comparison.

- **Space INPUT switches**



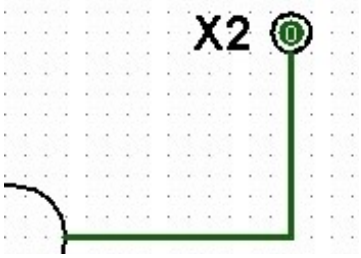
These switches represent the user-entered password. The user inputs a 4-bit value, and the circuit checks whether this value matches the stored password.

- **X1 — valid password indicator**



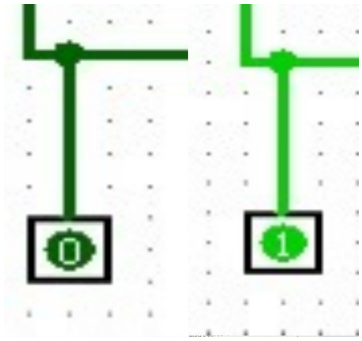
X1 is the output that turns ON when the entered password is correct. It becomes 1 (high) only when all four bits match between the stored password and the input password.

- **X2 — invalid password indicator**



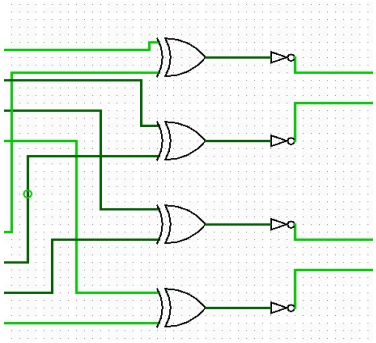
X2 is the output that turns ON when the entered password is incorrect. It activates when any of the bits does not match.

- **Enable switch (system activation input)**



This input works as a system enable switch. When this switch equals 1, the system becomes active and the AND gate allows the comparison results to appear on X1 or X2. If it equals 0, the lock system stays disabled, and no output is produced.

- **XOR Gates**



Each XOR gate is used to compare one bit of the stored password with the corresponding bit of the user input, this allows the circuit to check each bit pair independently. To know whether the entire 4-bit password is correct, the circuit must first confirm that each individual bit is the same by comparing bit-by-bit. XOR outputs 1

when the bits are different, and outputs 0 when the bits are the same. After connecting each output of the XOR gates to the NOT gates, they act as an XNOR gates, therefore the signal becomes 1 if the bits match, and 0 if they don't match. So every XOR receives one wire from the stored password (STORE) and one wire from the user input (INPUT), which allows the circuit to check each bit pair independently and create a "match signal" for each bit.

Chapter Three

System Logic Operation + Final Output Behavior

This part explains how the circuit processes the 4-bit password and how the outputs (X1 and X2) are generated based on the user input. The system uses a combination of XOR, NOT (XNOR), and AND gates to determine whether the entered password matches the stored password.

1. Bit-by-Bit Comparison Using XOR and NOT (XNOR)

Each bit from the stored password is connected to the matching bit from the user input through an XOR gate.
The XOR gate outputs:

- **1** when the two bits are different
- **0** when the two bits are equal

However, the system needs a signal that becomes **1 when both bits are the same**, so a NOT gate is added after every XOR gate. This converts the XOR output into an **XNOR output**, meaning:

- If the two bits match → XNOR = **1**
- If the two bits do not match → XNOR = **0**

This gives the system four "match signals," one for each bit of the password.

2. Using AND Gates to Confirm Full Password Match

After obtaining four XNOR outputs, the circuit checks whether *all four bits* match.
The four outputs are grouped and passed through **two AND gates**, so that:

- If even one bit is incorrect → one XNOR becomes 0 → the AND output becomes **0**
- If all bits match → all XNOR outputs are 1 → the AND output becomes **1**

This combined AND output represents the **overall password correctness**.

3. Final Output Generation (X1 and X2) Using Enable Input

The output of the final AND gate is then used to control the two indicators:

• **Valid Password Output (X1)**

The AND result is connected directly to another AND gate along with the **Enable switch**.
Therefore:

- If password is correct (1) **AND** enable = 1 → X1 becomes **1**
- Otherwise → X1 stays **0**

This means **X1 turns ON only when the password is correct and the system is enabled.**

• **Invalid Password Output (X2)**

To generate X2, the same AND result is first passed through a NOT gate.
So:

- Correct password → AND = 1 → NOT = 0 → X2 stays off

- Wrong password → AND = 0 → NOT = 1 → X2 turns on

Then the NOT output is passed through another AND gate with the **Enable switch**, ensuring the system works only when enabled.

4. Final Behavior Summary

- **If the password is correct:**
  - X1 = 1 (valid password indicator)
  - X2 = 0
- **If the password is incorrect:**
  - X1 = 0
  - X2 = 1
- **If Enable = 0:**
  - Both X1 and X2 remain 0 regardless of input

This completes the full logic operation of the simple 4-bit password security system.

Conclusion

In this project, a complete digital password-lock system was designed using only basic logic gates without any microcontroller or programming. The design successfully demonstrated how XOR, NOT (XNOR), and AND gates can be combined to build a functional 4-bit password checker. By comparing each bit individually using XNOR logic and then verifying all bits together using AND gates, the system ensures accurate password matching.

The addition of an enable switch allows the circuit to activate only when desired, making the system more practical and secure. The final outputs clearly indicate whether the entered password is valid or invalid. Overall, this project helped us understand digital comparison, decision-making in hardware, and how complex functions can be implemented using simple logic gates.

References

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